

Nonlinear latent manifolds and closure models for reduced order models

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Abstract

We compare four ANN-enhanced reduced-order modeling routes under aggressive online compression: (i) closure-based intrusive ROMs (PROM-ANN), (ii) latent-manifold intrusive ROMs (PROM-POD-AE), (iii) non-intrusive coefficient-space POD-NN-ROM surrogates $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$, and (iv) non-intrusive latent POD-DL-ROM surrogates $(\boldsymbol{\mu}, t) \mapsto \mathbf{z} \mapsto \mathbf{q}_N$. For intrusive models, the online state is obtained by time-discrete least-squares Petrov–Galerkin (LSPG) minimization, solved by Gauss–Newton, and accelerated by ECSW hyperreduction. The closure family includes state-conditioned, parameter–time conditioned, and hybrid maps for truncated coefficients. The latent family evolves low-dimensional latent coordinates and reconstructs reduced coefficients through a decoder.

A central methodological choice is ROM-consistent training: learning targets are generated by a linear n_{tot} -dimensional HPROM solve in the full truncated space, rather than by direct projection of HDM trajectories. We also consider a second enrichment stage built from additional low-cost HPROM trajectories at Latin-hypercube parameter points. This manuscript consolidates the mathematical formulation and experimental protocol for the 250×250 2D parametric inviscid Burgers benchmark and prepares an extension to a shallow-water benchmark.

1 Introduction

Projection-based model order reduction (PMOR) is a standard framework for reducing high-dimensional parametric models while preserving governing equations [1, 2]. For nonlinear transport-dominated dynamics, linear subspaces may require large dimensions, which motivates nonlinear manifolds, closure modeling, and latent representations [3, 4, 5].

From a machine-learning viewpoint, four families are relevant to this work:

- intrusive closure-enhanced projection (PROM-ANN) [3],
- intrusive latent-manifold projection (PROM-POD-AE), in the spirit of hyperreduced nonlinear manifold ROMs [6],
- non-intrusive direct-coefficient regression maps $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$,
- non-intrusive latent POD-DL-ROM maps [7].

Naming convention used in this manuscript. When we say **POD-NN-ROM**, we refer to the non-intrusive coefficient-space map $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$. When we say **POD-DL-ROM**, we refer to the non-intrusive latent family in the sense of Fresca–Manzoni [7]. When we say **PROM-POD-AE**, we refer to an intrusive latent-manifold projection route (conceptually aligned with intrusive nonlinear-manifold HPROM literature such as [6]), even though those references use different method names.

The goal is a consistent comparison under strong online compression, using the same HDM residual, same time discretization, and the same HPROM intrusive solver/hyperreduction stack for intrusive variants.

2 Linear HPROM Reference and Notation

2.1 Generic semi-discrete IVP and residual notation

Given the generic, parametric, nonlinear, semi-discrete, first-order initial value problem

$$\mathbf{M}(\boldsymbol{\mu}) \dot{\mathbf{u}}(t; \boldsymbol{\mu}) + \mathbf{f}(\mathbf{u}(t; \boldsymbol{\mu}); \boldsymbol{\mu}) - \mathbf{g}(t; \boldsymbol{\mu}) = \mathbf{0}, \quad \mathbf{u}(0; \boldsymbol{\mu}) = \mathbf{u}_0(\boldsymbol{\mu}), \quad (1)$$

with $\boldsymbol{\mu} \in \mathcal{D} \subset \mathbb{R}^p$, we denote by $\mathbf{r}^{m+1}$ the nonlinear time-discrete residual associated with the selected implicit time integrator. In compact form, each HDM step is written as

$$\mathbf{r}^{m+1}(\mathbf{u}^{m+1}; \boldsymbol{\mu}) = \mathbf{0}. \quad (2)$$

For readability, dependencies on previous time states and Δt are kept implicit in $\mathbf{r}^{m+1}$.

2.2 POD basis from SVD

Let $\mathbf{u}(t; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the HDM state with parameters $\boldsymbol{\mu} \in \mathcal{D} \subset \mathbb{R}^p$. Define

$$\mathbf{u}_{\text{ref}} := \frac{1}{N_s} \sum_{i=1}^{N_s} \mathbf{u}^i, \quad \mathbf{S}_c = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}} \ \cdots \ \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}]. \quad (3)$$

Let

$$\mathbf{S}_c = \mathbf{U}_S \boldsymbol{\Sigma}_S \mathbf{Y}_S^T \quad (4)$$

be its thin SVD, and define the linear ROB

$$\mathbf{V}_{\text{tot}} \in \mathbb{R}^{N \times n_{\text{tot}}} \quad (5)$$

as the first n_{tot} left singular vectors of $\mathbf{U}_S$.

The linear reduced reconstruction is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N(t; \boldsymbol{\mu}), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \in \mathbb{R}^{n_{\text{tot}}}. \quad (6)$$

2.3 Linear time-discrete LSPG reference form

Let $\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the HDM time-discrete residual at step $m+1$. The unsampled reference linear LSPG solve is

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (7)$$

With

$$\mathbf{J}^{m+1,k} := \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}}, \quad \mathbf{T}_N := \mathbf{V}_{\text{tot}}, \quad \mathbf{W}_N^{m+1,k} := \mathbf{J}^{m+1,k} \mathbf{T}_N, \quad (8)$$

the Gauss–Newton increment solves

$$(\mathbf{W}_N^{m+1,k})^T \mathbf{W}_N^{m+1,k} \Delta \mathbf{q}_N^{m+1,k} = -(\mathbf{W}_N^{m+1,k})^T \mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N^{m+1,k}; \boldsymbol{\mu}). \quad (9)$$

2.4 Linear HPROM via ECSW

Assume element-wise residual decomposition

$$\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) = \sum_{e_i \in \mathcal{E}} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i}^+ \mathbf{u}; \boldsymbol{\mu}). \quad (10)$$

ECSW approximation yields

$$\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu}) = \sum_{e_i \in \tilde{\mathcal{E}}} \xi_{e_i} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i}^+ [\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{x}]; \boldsymbol{\mu}), \quad \xi_{e_i} > 0, \quad (11)$$

with sampled set $\tilde{\mathcal{E}} \subset \mathcal{E}$ [8, 9]. The linear HPROM solve is

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (12)$$

ROM-consistent training targets. All learning targets in this work are extracted from trajectories $\mathbf{q}_N(t; \boldsymbol{\mu})$ generated by the linear HPROM solve (12). Equation (7) is retained only as the unsampled LSPG reference form.

3 Method I: PROM-ANN Closure Models

3.1 Manifold definitions

For closure models, we introduce the partition

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \overline{\mathbf{V}}], \quad \mathbf{q}_N = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}, \quad n + \bar{n} = n_{\text{tot}}. \quad (13)$$

PROM-ANN uses retained coordinates $\mathbf{q} \in \mathbb{R}^n$ as online unknowns and reconstructs truncated coordinates via learned maps:

$$\text{Case 1: } \bar{\mathbf{q}} = \mathcal{N}(\mathbf{q}), \quad (14)$$

$$\text{Case 2: } \bar{\mathbf{q}} = \mathcal{M}(\boldsymbol{\mu}, t), \quad (15)$$

$$\text{Case 3: } \bar{\mathbf{q}} = \mathcal{H}(\mathbf{q}, \boldsymbol{\mu}, t). \quad (16)$$

The state approximation is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}(t; \boldsymbol{\mu}) + \overline{\mathbf{V}} \mathcal{F}(\mathbf{q}, \boldsymbol{\mu}, t), \quad (17)$$

with $\mathcal{F} \in \{\mathcal{N}, \mathcal{M}, \mathcal{H}\}$.

3.2 LSPG/Gauss–Newton solve

At each time step,

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{x}; \boldsymbol{\mu}); \boldsymbol{\mu})\|_2^2. \quad (18)$$

Define

$$\mathbf{J}^{m+1,k} := \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}}, \quad \mathbf{T}_q^{m+1,k} := \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{q}} = \mathbf{V} + \overline{\mathbf{V}} \frac{\partial \mathcal{F}}{\partial \mathbf{q}}, \quad \mathbf{W}_q^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}_q^{m+1,k}. \quad (19)$$

The Gauss–Newton increment solves

$$(\mathbf{W}_q^{m+1,k})^T \mathbf{W}_q^{m+1,k} \Delta \mathbf{q}^{m+1,k} = -(\mathbf{W}_q^{m+1,k})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}). \quad (20)$$

3.3 ECSW hyperreduced form

PROM-ANN HPROM solves

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2, \quad (21)$$

with test basis $\widetilde{\mathbf{W}}_q = \widetilde{\mathbf{J}} \mathbf{T}_q$.

Offline training targets. For Cases 1–3, supervised data are extracted from ROM-consistent trajectories $\mathbf{q}_N = [\mathbf{q}; \bar{\mathbf{q}}]$. Case 1 trains $\mathcal{N}$ from $(\mathbf{q}, \bar{\mathbf{q}})$, Case 2 trains $\mathcal{M}$ from $(\boldsymbol{\mu}, t, \bar{\mathbf{q}})$, and Case 3 trains $\mathcal{H}$ from $(\mathbf{q}, \boldsymbol{\mu}, t, \bar{\mathbf{q}})$.

4 Method II: PROM-POD-AE Latent Manifold

4.1 Autoencoder and latent manifold

Let

$$\mathbf{z} = \mathcal{E}(\mathbf{q}_N), \quad \hat{\mathbf{q}}_N = \mathcal{D}(\mathbf{z}), \quad \mathbf{z} \in \mathbb{R}^{n_z}, \quad n_z \ll n_{\text{tot}}. \quad (22)$$

The intrusive latent manifold is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{z}(t; \boldsymbol{\mu})). \quad (23)$$

This is an intrusive latent-manifold projection approach, in line with intrusive nonlinear-manifold HPROM literature [6], and is separated from the non-intrusive POD-DL-ROM line [7].

Offline autoencoder training. Given ROM-consistent targets $\mathbf{q}_N^{\text{ref}}$, the autoencoder $(\mathcal{E}, \mathcal{D})$ is trained by reconstruction in reduced coordinates:

$$\mathcal{L}_{\text{AE}} = \frac{1}{N_{\text{samp}}} \sum_{j=1}^{N_{\text{samp}}} \left\| \mathbf{q}_{N,j}^{\text{ref}} - \mathcal{D}(\mathcal{E}(\mathbf{q}_{N,j}^{\text{ref}})) \right\|_2^2. \quad (24)$$

The resulting decoder $\mathcal{D}$ defines the nonlinear trial manifold used in the intrusive online LSPG solve (25).

4.2 LSPG/Gauss–Newton solve in latent coordinates

The online unknown is $\mathbf{z}^{m+1} \in \mathbb{R}^{n_z}$:

$$\mathbf{z}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{y} \in \mathbb{R}^{n_z}} \left\| \mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{y}); \boldsymbol{\mu}) \right\|_2^2. \quad (25)$$

Define

$$\mathbf{T}_z^{m+1,k} := \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{z}} = \mathbf{V}_{\text{tot}} \frac{\partial \mathcal{D}}{\partial \mathbf{z}}, \quad \mathbf{W}_z^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}_z^{m+1,k}. \quad (26)$$

The Gauss–Newton step solves

$$(\mathbf{W}_z^{m+1,k})^T \mathbf{W}_z^{m+1,k} \Delta \mathbf{z}^{m+1,k} = -(\mathbf{W}_z^{m+1,k})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{z}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}). \quad (27)$$

4.3 ECSW hyperreduced form

PROM-POD-AE HPROM solves

$$\mathbf{z}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{y} \in \mathbb{R}^{n_z}} \left\| \tilde{\mathbf{r}}^{m+1}(\mathbf{y}; \boldsymbol{\mu}) \right\|_2^2, \quad (28)$$

with test basis $\widetilde{\mathbf{W}}_z = \widetilde{\mathbf{J}} \mathbf{T}_z$.

5 Method III: POD-NN-ROM (Coefficient-Space, Non-Intrusive)

Model definition. The non-intrusive baseline maps exogenous inputs directly to reduced coefficients:

$$\mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{G}_q(\boldsymbol{\mu}, t). \quad (29)$$

This model does not require online residual minimization.

Training objective and online cost. Given ROM-consistent targets $\mathbf{q}_N^{\text{ref}}(t; \boldsymbol{\mu})$, one minimizes a supervised loss (e.g., MSE)

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{samp}}} \sum_{j=1}^{N_{\text{samp}}} \left\| \mathcal{G}_q(\boldsymbol{\mu}_j, t_j) - \mathbf{q}_{N,j}^{\text{ref}} \right\|_2^2. \quad (30)$$

Online prediction is then a direct forward evaluation of $\mathcal{G}_q$ followed by reconstruction with $\mathbf{V}_{\text{tot}}$.

6 Method IV: POD-DL-ROM (Latent, Non-Intrusive)

Model definition. The second non-intrusive baseline follows the POD-DL-ROM family [7]:

$$\mathbf{z}(t; \boldsymbol{\mu}) \approx \mathcal{G}_z(\boldsymbol{\mu}, t), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{D}(\mathcal{G}_z(\boldsymbol{\mu}, t)), \quad (31)$$

where $\mathcal{D}$ is a decoder trained offline and $\mathbf{z} \in \mathbb{R}^{n_z}$.

Training objective and online cost. In the present formulation, the autoencoder block $(\mathcal{E}, \mathcal{D})$ follows the same training strategy and normalization used in Section 4: reconstruction of ROM-consistent $\mathbf{q}_N$ snapshots through a latent bottleneck. The POD-DL-ROM dynamics map then replaces the POD-NN-ROM output variable $\mathbf{q}_N$ by $\mathbf{z}$, i.e., it learns $(\boldsymbol{\mu}, t) \mapsto \mathbf{z}$ and composes with the same decoder $\mathcal{D}$:

$$\mathbf{q}_N(\boldsymbol{\mu}, t) \approx \mathcal{D}(\mathcal{G}_z(\boldsymbol{\mu}, t)). \quad (32)$$

Hence, POD-NN-ROM and POD-DL-ROM share the same exogenous-input principle $(\boldsymbol{\mu}, t) \mapsto (\cdot)$, with POD-DL-ROM acting in latent coordinates as proposed in [7]. Online evaluation remains non-intrusive (single forward pass) and does not involve residual minimization.

7 Physics-Regularized Variant (Ancillary)

Physics-regularized neural training adds a residual-based term to a supervised objective [10, 11]. In this study, such variants are treated as optional diagnostics (not core baselines) because they remain significantly more expensive offline and do not consistently improve accuracy in the present benchmark.

8 Burgers Benchmark and Experimental Protocol

8.1 Governing equations and parameter domain

We use the 2D parametric inviscid Burgers benchmark from [3], in conservative vector form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad (33)$$

with

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2} u_x^2 \\ \frac{1}{2} u_x u_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2} u_x u_y \\ \frac{1}{2} u_y^2 \end{bmatrix}, \quad \mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}. \quad (34)$$

The problem is defined on $\Omega = [0, 100] \times [0, 100]$ with final time $T_f = 25$ and parameters $\boldsymbol{\mu} = (\mu_1, \mu_2) \in \mathcal{D}$, where

$$\mathcal{D} = [4.25, 5.50] \times [0.015, 0.03]. \quad (35)$$

This conservative form is the one used by the discrete residual and Jacobian in the ROM/HPROM pipeline. The initial/boundary conditions used in this benchmark are

$$u_x(0, y, t; \boldsymbol{\mu}) = \mu_1, \quad u_x(x, y, 0) = 1, \quad u_y(x, y, 0) = 1. \quad (36)$$

8.2 Numerical discretization

The HDM uses a cell-centered finite-volume Godunov discretization on a 250×250 mesh, hence $N = 2N_x N_y = 125,000$. Time integration uses 500 uniform steps with $\Delta t = 0.05$ (thus 501 stored states including $t = 0$). The same time-discrete residual is used by linear HPROM and by all intrusive ANN-enhanced variants (PROM-ANN and PROM-POD-AE) through LSPG/Gauss–Newton solves.

8.3 Training and evaluation protocol

Stage 1 builds $\mathbf{V}_{\text{tot}}$ from HDM snapshots at 9 structured training parameters (a 3×3 tensor grid on $\mathcal{D}$), giving $N_s = 9 \times 501 = 4509$ snapshots. Stage 2 generates ROM-consistent coefficients $\mathbf{q}_N(t; \boldsymbol{\mu})$ by solving the linear full-space reduced model (n_{tot}). Stage 3 trains the learned maps from these reduced targets.

Evaluation is reported at one verification point and two off-grid points:

$$\boldsymbol{\mu}^{(v)} = (4.875, 0.0225), \quad \boldsymbol{\mu}^{(1)} = (4.56, 0.019), \quad \boldsymbol{\mu}^{(2)} = (5.19, 0.026). \quad (37)$$

An enrichment stage adds 20 Latin-hypercube points in parameter space. The enrichment data is generated with reduced solves (no additional HDM runs), then the same model families are retrained and re-evaluated at the same three points. Figure 1 summarizes the baseline, enrichment, and evaluation locations in $\mathcal{D}$.

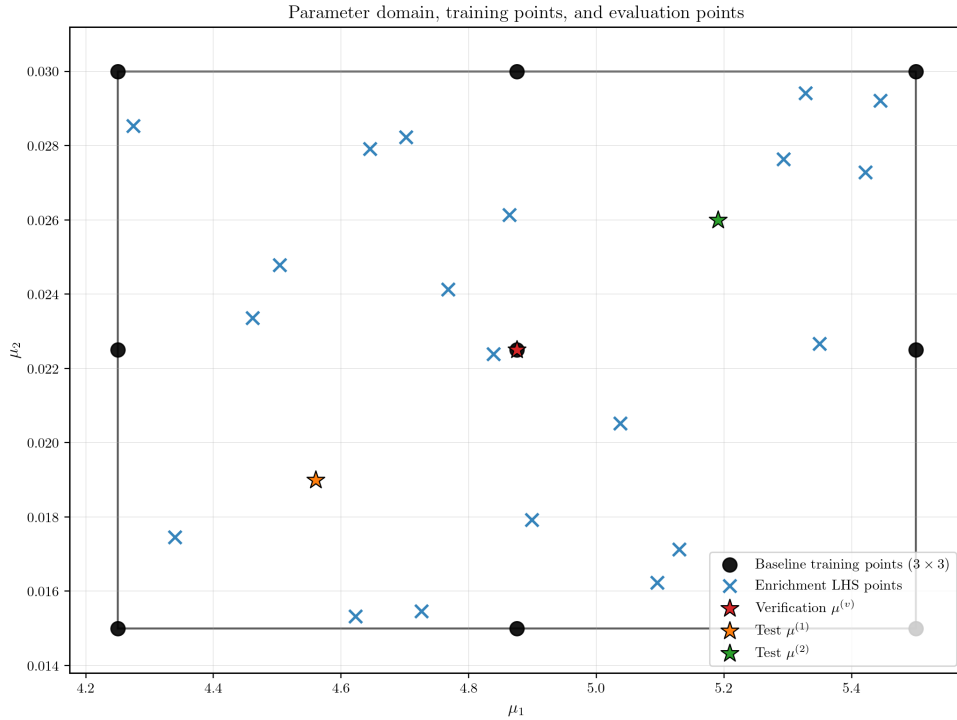


Figure 1: Parameter domain $\mathcal{D}$, baseline 3×3 training points, enrichment LHS points, and evaluation points $\boldsymbol{\mu}^{(v)}$, $\boldsymbol{\mu}^{(1)}$, $\boldsymbol{\mu}^{(2)}$.

8.4 HPROM-Consistent Evaluation Pipeline

To avoid protocol-induced bias in intrusive comparisons, we enforce one end-to-end HPROM pipeline: Stage 2 reduced targets are generated with linear HPROM in $\mathbf{V}_{\text{tot}}$, Stage 3 intrusive

models are trained on these HPROM-consistent reduced trajectories, and online intrusive evaluations are performed with linear HPROM settings. This applies uniformly to Linear HPROM, PROM-ANN (Cases 1–3), and PROM-POD-AE.

8.5 Reported Metrics

For each evaluation parameter, we report complementary state-space, coefficient-space, and runtime metrics. State accuracy is measured by the relative space-time trajectory error versus HDM (defined in Section 9, Error Definitions). Coefficient-space comparisons are reported against the linear n_{tot} reference, including per-index absolute/relative modal errors and time-resolved modal error heatmaps. Computational performance is reported through online runtimes and speedups relative to the linear HPROM reference.

9 Burgers Results in the HPROM-Consistent Setting

This section reports the Burgers results for the six model families considered: linear HPROM reference, PROM-ANN Cases 1–3, intrusive PROM-POD-AE, non-intrusive POD-NN-ROM $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$, and non-intrusive POD-DL-ROM.

9.1 Error Definitions

The trajectory error reported in tables and summaries is the relative space-time Frobenius error (in %):

$$\varepsilon_u^{(m)}(\boldsymbol{\mu}) = 100 \frac{\|\mathbf{U}_{\text{HDM}}(\boldsymbol{\mu}) - \mathbf{U}_m(\boldsymbol{\mu})\|_F}{\|\mathbf{U}_{\text{HDM}}(\boldsymbol{\mu})\|_F}, \quad (38)$$

where $\mathbf{U}_{\text{HDM}}, \mathbf{U}_m \in \mathbb{R}^{N \times N_t}$ collect all states along time.

Coefficient diagnostics use the linear n_{tot} -HPROM coefficients $\mathbf{Q}_{\text{ref}} = [\mathbf{q}_{\text{ref}}(t_1) \cdots \mathbf{q}_{\text{ref}}(t_{N_t})]$ as reference and model coefficients $\mathbf{Q}_m$:

$$\varepsilon_{q_i, \text{abs}}^{(m)} = \|q_{\text{ref}, i}(\cdot) - q_{m, i}(\cdot)\|_2, \quad \varepsilon_{q_i, \text{rel}}^{(m)} = \frac{\|q_{\text{ref}, i}(\cdot) - q_{m, i}(\cdot)\|_2}{\|q_{\text{ref}, i}(\cdot)\|_2}, \quad (39)$$

and heatmaps display

$$e_{i, k}^{(m)} = |q_{\text{ref}, i}(t_k) - q_{m, i}(t_k)|, \quad \hat{e}_{i, k}^{(m)} = \frac{|q_{\text{ref}, i}(t_k) - q_{m, i}(t_k)|}{\|q_{\text{ref}, i}(\cdot)\|_2}. \quad (40)$$

For online cost tables, intrusive methods are timed with their online Gauss–Newton solve, while non-intrusive methods are timed through pure network inference. Speedup is defined as

$$S^{(m)} = \frac{t_{\text{Linear}}}{t_m}. \quad (41)$$

9.2 Trajectory Errors and Online Cost

Table 1 reports baseline relative trajectory errors (in %) at the three evaluation points, and Table 2 reports the enriched counterpart. Enrichment improves all learned models, with the strongest correction in Case 2 at the off-grid points. For reproducibility, the baseline POD-DL-ROM row corresponds to a trained checkpoint with latent size $n_z = 20$, while the enriched POD-DL-ROM row uses the enriched checkpoint with $n_z = 5$.

Table 1: Baseline stage, HPROM online solve: relative trajectory errors (%) vs HDM.

Point	Linear HPROM	PROM-ANN Case 1	PROM-ANN Case 2	PROM-ANN Case 3	PROM-POD-AE	POD-NN-ROM	POD-DL-ROM
$\boldsymbol{\mu}^{(v)}$	0.410	0.620	1.239	0.781	0.785	0.836	0.988
$\boldsymbol{\mu}^{(1)}$	0.450	0.823	2.872	0.778	0.868	1.886	1.945
$\boldsymbol{\mu}^{(2)}$	0.463	1.014	1.501	1.022	1.528	1.667	2.967

Table 2: Enriched stage, HPROM online solve: relative trajectory errors (%) vs HDM.

Point	Linear HPROM	PROM-ANN Case 1	PROM-ANN Case 2	PROM-ANN Case 3	PROM-POD-AE	POD-NN-ROM	POD-DL-ROM
$\mu^{(v)}$	0.410	0.563	0.649	0.493	0.585	0.710	0.752
$\mu^{(1)}$	0.450	0.614	0.927	0.606	0.718	0.806	0.831
$\mu^{(2)}$	0.463	0.885	0.614	0.550	1.033	0.784	0.822

Average online runtimes and speedups are summarized in Table 3. As expected, intrusive reduced solves remain one to two orders of magnitude faster than linear HPROM, while the non-intrusive direct and latent data-driven maps are effectively instantaneous online.

Table 3: Average online cost over $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$.

Method	Baseline time (s)	Enriched time (s)	Baseline speedup	Enriched speedup
Linear HPROM	136.63	136.63	1.00	1.00
PROM-ANN Case 1	16.13	16.53	8.48	8.27
PROM-ANN Case 2	5.62	5.76	24.34	23.75
PROM-ANN Case 3	18.43	18.22	7.42	7.54
PROM-POD-AE	12.70	13.21	10.82	10.45
POD-NN-ROM	0.0052	0.0053	2.64×10^4	2.57×10^4
POD-DL-ROM	0.0086	0.0047	1.59×10^4	2.89×10^4

9.3 Solution Overlays

Figures 2 and 3 show the HDM verification snapshot in 3D at two times, together with the two cut-planes used by the 1D line comparisons.

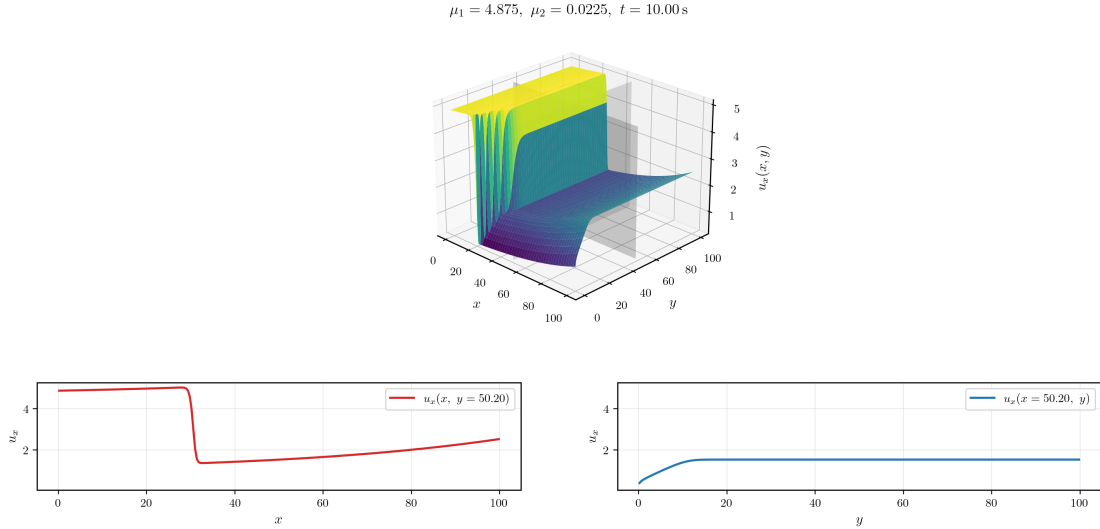


Figure 2: Verification-point HDM snapshot at $t = 10 \text{ s}$ (top) and extracted cut-planes (bottom).

$$\mu_1 = 4.875, \mu_2 = 0.0225, t = 20.00 \text{ s}$$

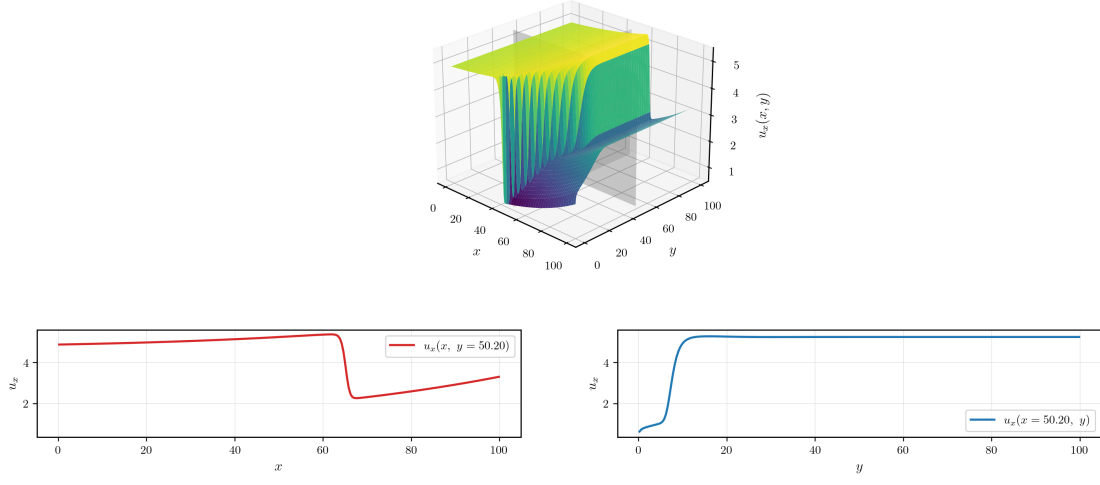


Figure 3: Verification-point HDM snapshot at $t = 20$ s (top) and extracted cut-planes (bottom).

Figures 4 and 5 compare HDM and all considered models at $\mu^{(v)}$, $\mu^{(1)}$, and $\mu^{(2)}$.

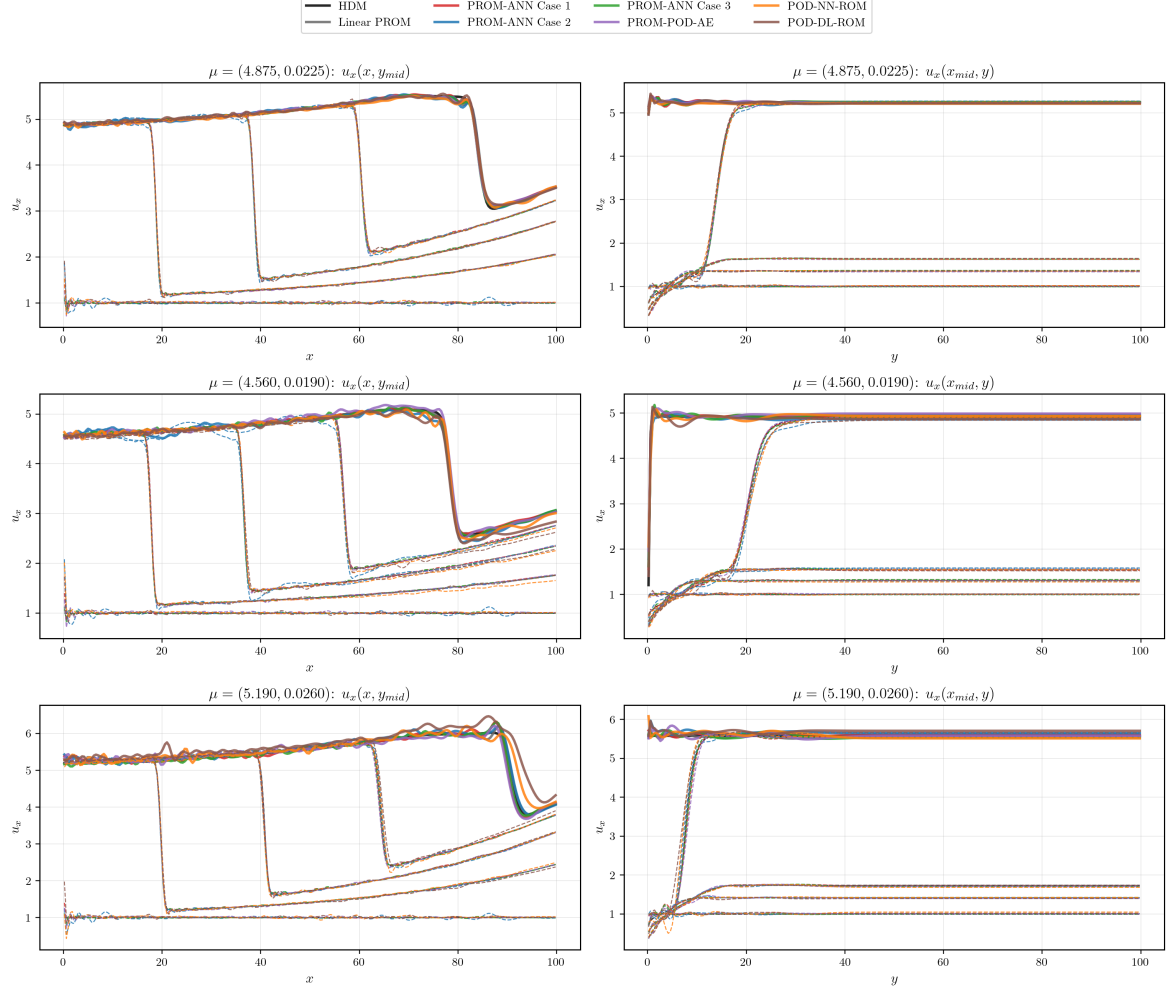


Figure 4: Baseline models, HPROM online solve: HDM vs Linear HPROM, PROM-ANN Case 1, PROM-ANN Case 2, PROM-ANN Case 3, PROM-POD-AE, POD-NN-ROM, and POD-DL-ROM at the three evaluation points.

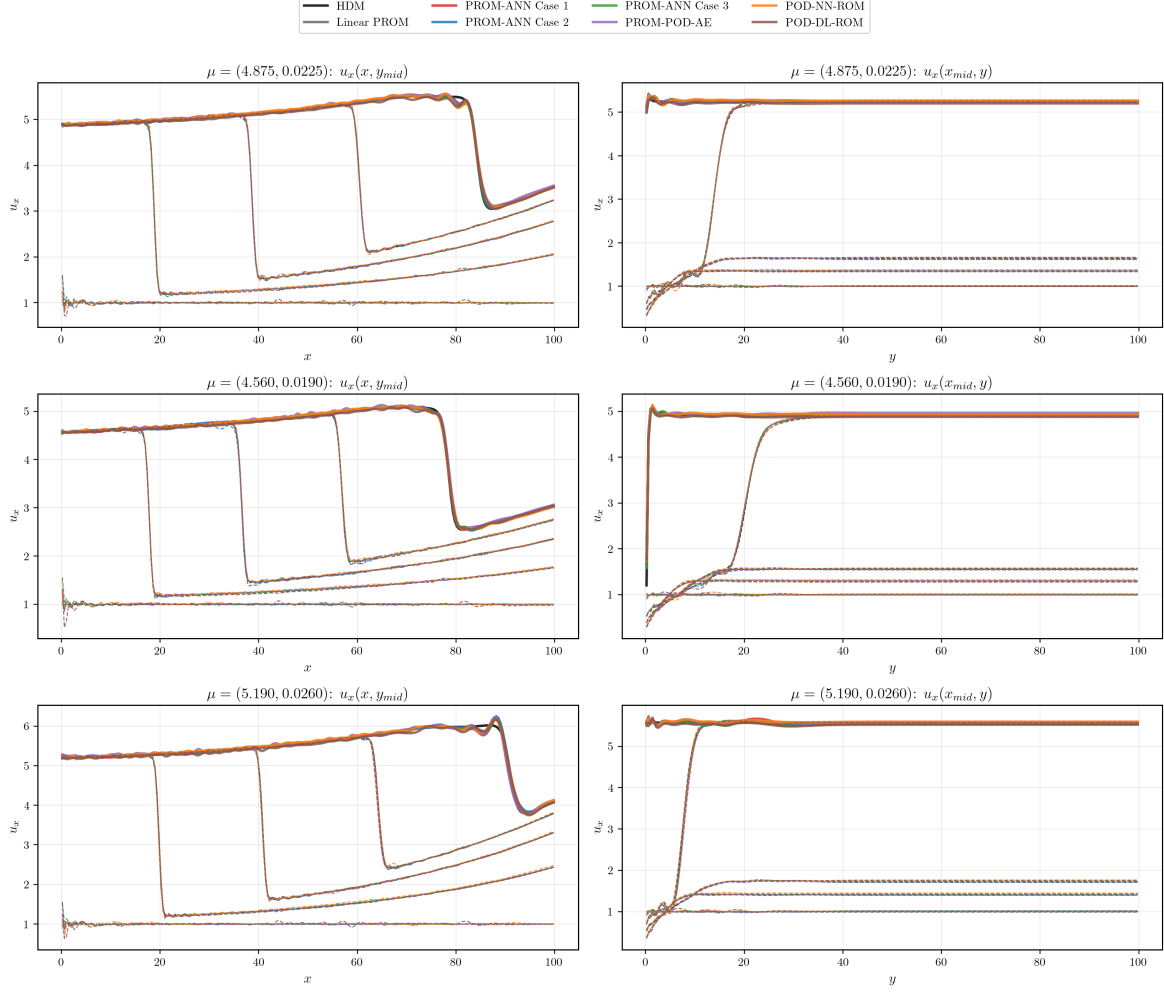


Figure 5: Enriched models, HPROM online solve: HDM vs Linear HPROM, PROM-ANN Case 1, PROM-ANN Case 2, PROM-ANN Case 3, PROM-POD-AE, POD-NN-ROM, and POD-DL-ROM at the three evaluation points.

9.4 Coefficient-Space Diagnostics

We use the linear n_{tot} -HPROM coefficients as reference and report per-index absolute/relative coefficient errors and time-resolved heatmaps for all learned models. Figures 6 and 7 show per-index curves at the verification point, while Figures 8 and 9 show the corresponding absolute-error heatmaps.

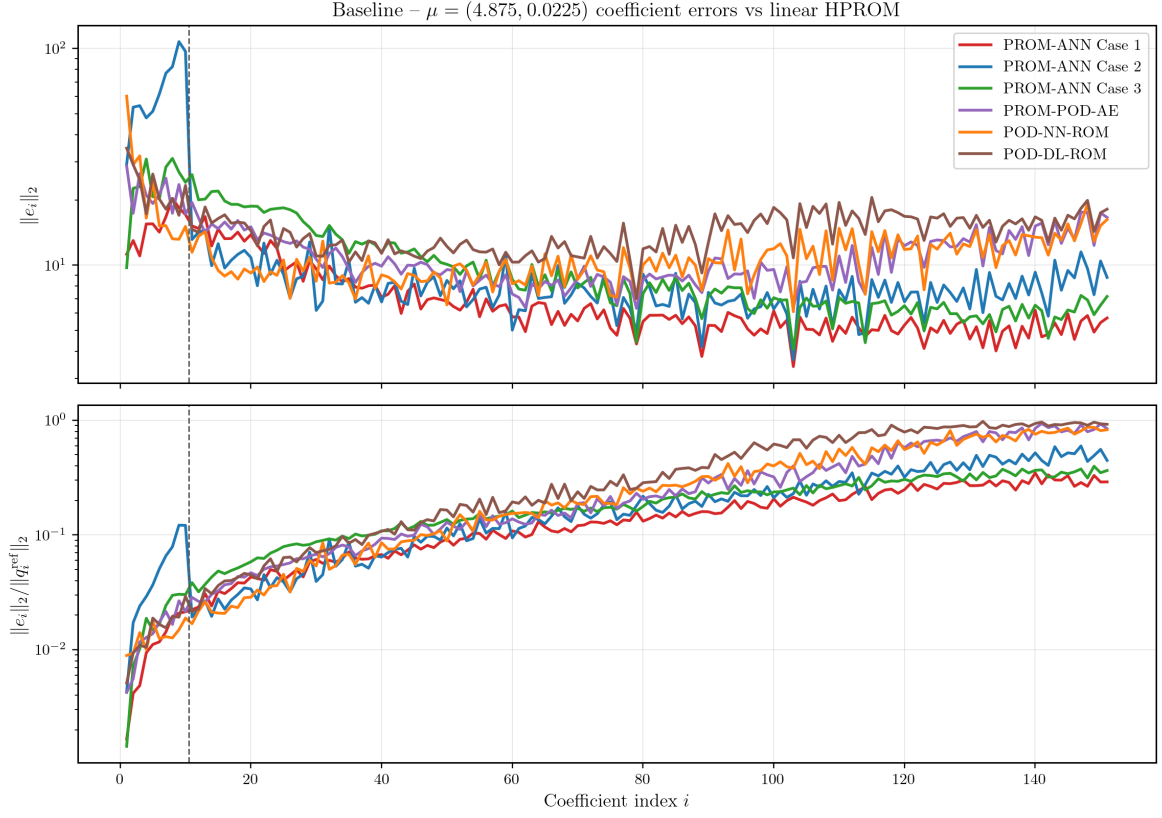


Figure 6: Baseline stage: per-index absolute and relative coefficient errors at $\boldsymbol{\mu}^{(v)} = (4.875, 0.0225)$.

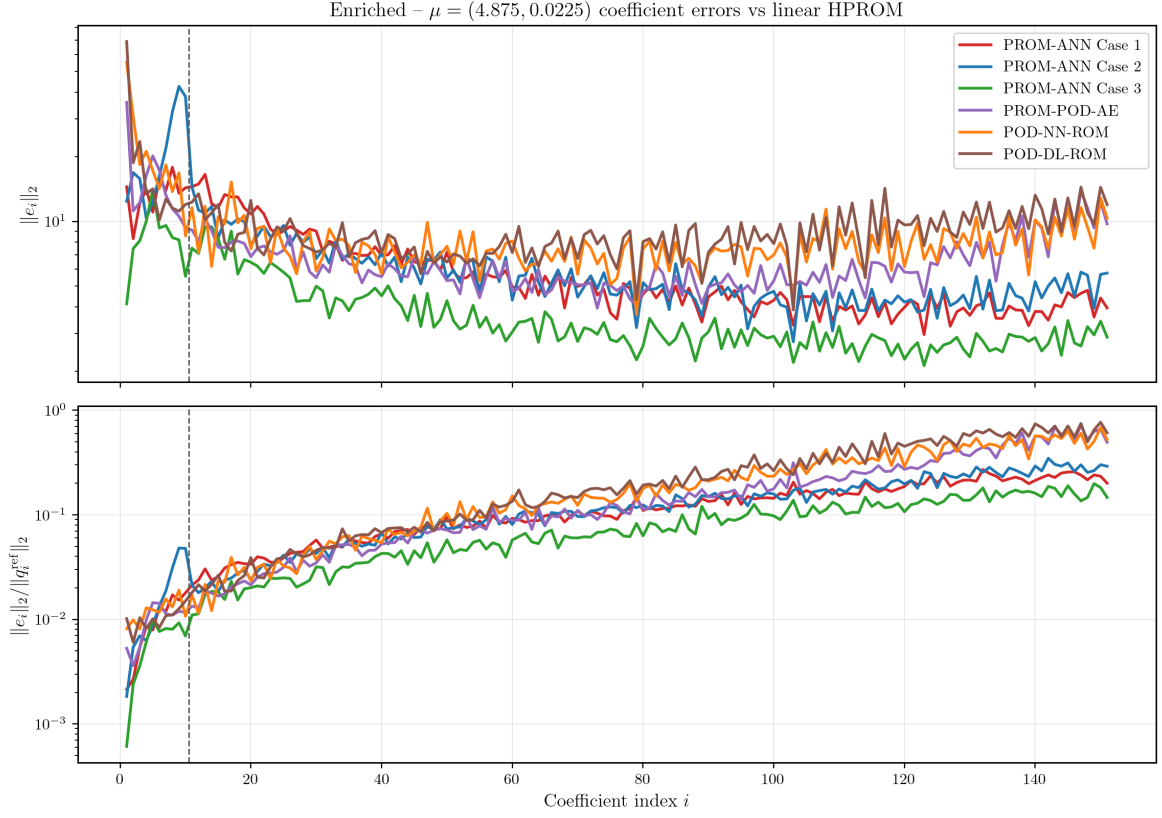


Figure 7: Enriched stage: per-index absolute and relative coefficient errors at $\mu^{(v)} = (4.875, 0.0225)$.

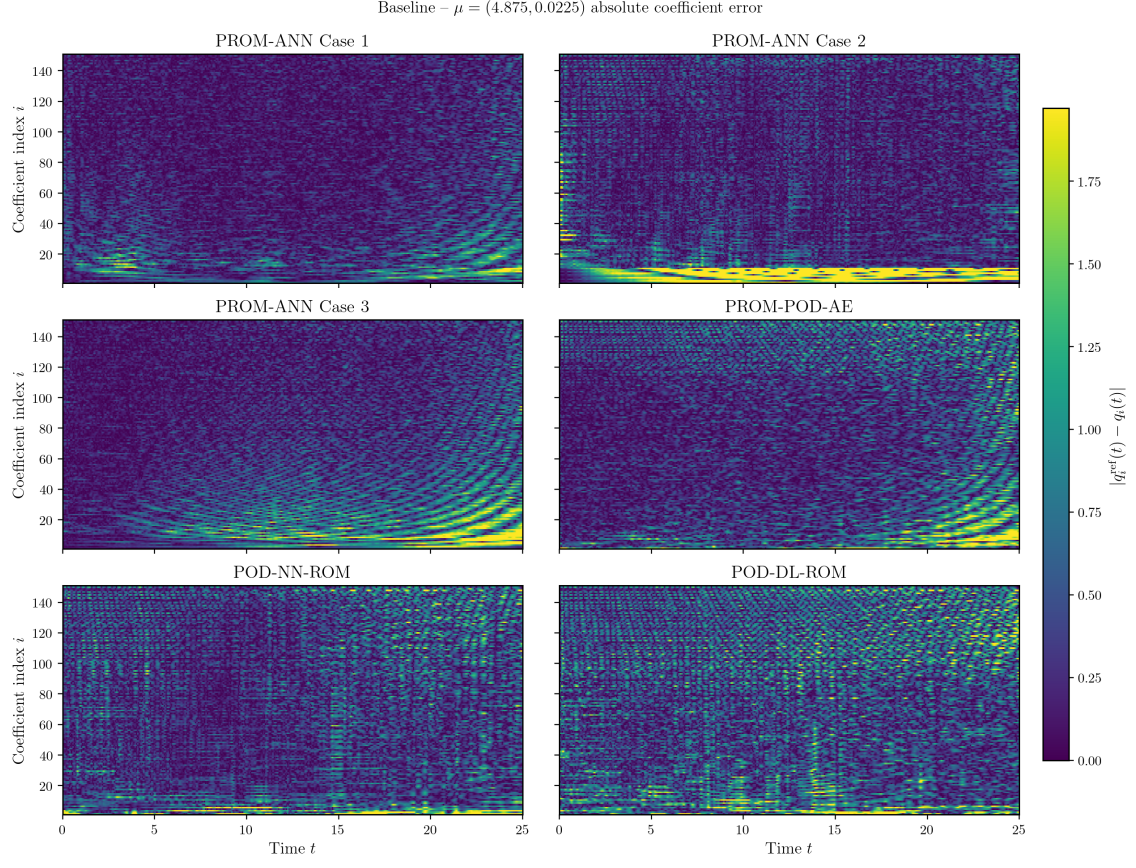


Figure 8: Baseline stage: absolute coefficient-error heatmaps at $\boldsymbol{\mu}^{(v)} = (4.875, 0.0225)$.

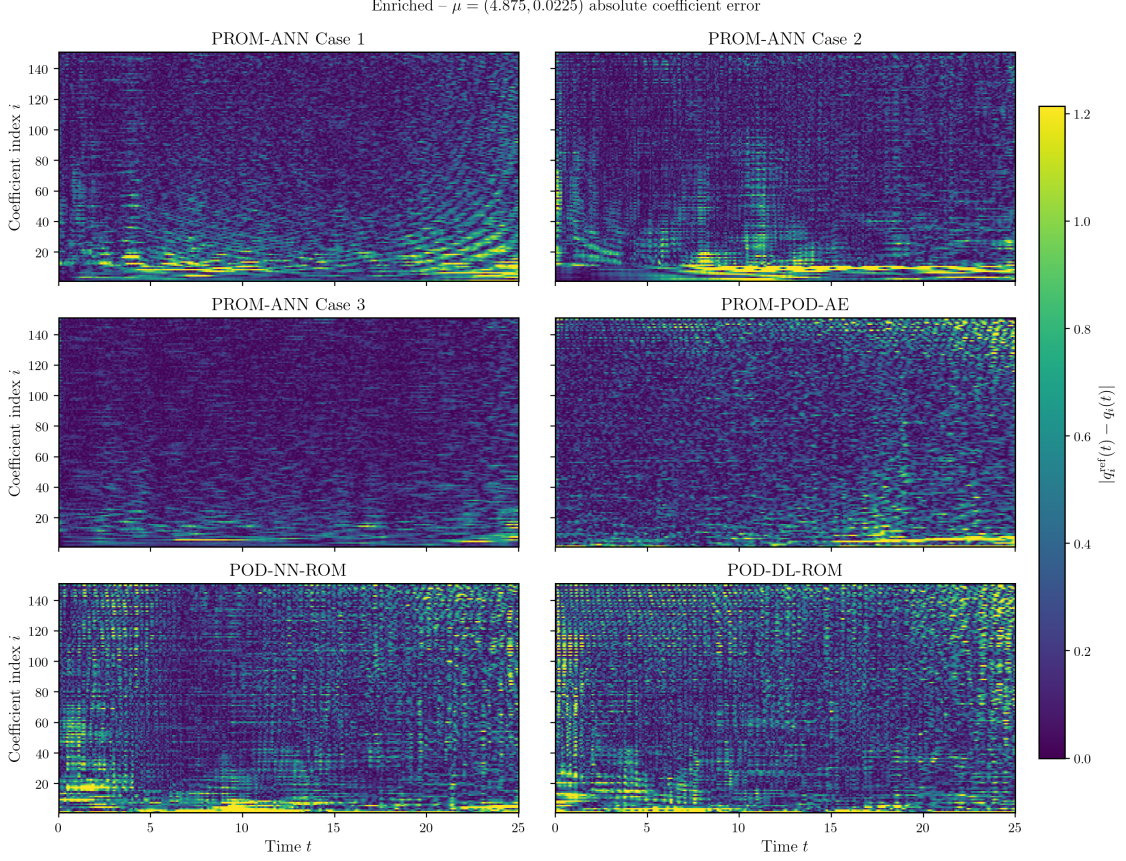


Figure 9: Enriched stage: absolute coefficient-error heatmaps at $\mu^{(v)} = (4.875, 0.0225)$.

Equivalent coefficient curves and heatmaps were generated for $\mu^{(1)} = (4.56, 0.019)$ and $\mu^{(2)} = (5.19, 0.026)$ with the same postprocessing settings.

10 Shallow-Water Benchmark and Experimental Protocol

10.1 Governing equations and parameterized initial condition

The second benchmark is a 2D shallow-water model on a flat bottom, in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \quad \mathbf{U} = \begin{bmatrix} h \\ hu_x \\ hu_y \end{bmatrix}, \quad (42)$$

with

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} hu_x \\ hu_x^2 + \frac{1}{2}gh^2 \\ hu_xu_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} hu_y \\ hu_xu_y \\ hu_y^2 + \frac{1}{2}gh^2 \end{bmatrix}. \quad (43)$$

The parametric scenario is a collision of two Gaussian bumps in water depth:

$$h(x, y, 0; \mu) = h_0 + \mu_1 \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_1\|_2^2}{2\sigma^2}\right) + \mu_2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_2\|_2^2}{2\sigma^2}\right), \quad (44)$$

with zero initial velocity components ($u_x = u_y = 0$), $h_0 = 1$, $\sigma = 0.045$, $\mathbf{c}_1 = (0.30, 0.50)$, $\mathbf{c}_2 = (0.70, 0.50)$, and

$$\mu = (\mu_1, \mu_2) \in [0.06, 0.14] \times [0.06, 0.14]. \quad (45)$$

Reflective boundary conditions are imposed on all sides.

10.2 Numerical setup

The domain $\Omega = [0, 1] \times [0, 1]$ is discretized with $N_x = N_y = 600$, leading to $N = 3N_xN_y$ unknowns. The HDM uses MUSCL reconstruction with HLL/HLLC interface fluxes and an implicit BDF2 time integrator (BE startup). We use a fixed time step $\Delta t = 7 \times 10^{-5}$ up to $T_f = 0.28$, yielding 4001 stored states per parameter value.

10.3 Sampling protocol aligned with Burgers

The staged workflow mirrors the Burgers protocol: baseline set of 9 structured parameter points (3×3 grid), enrichment set of 20 LHS points, Stage 1 POD basis $\mathbf{V}_{\text{tot}}$ from HDM snapshots, Stage 2 ROM-consistent $\mathbf{q}_N$ dataset from linear HPROM in $\mathbf{V}_{\text{tot}}$, Stage 3 training of PROM-ANN (Cases 1–3), PROM-POD-AE, POD-NN-ROM, and POD-DL-ROM, followed by online intrusive evaluation with HPROM settings. This protocol preserves methodological consistency across both benchmarks, so that observed performance differences are attributable to model structure and problem physics rather than to workflow changes.

11 Scope of the Study

Core model families are the linear HPROM reference, PROM-ANN (Cases 1–3), PROM-POD-AE, POD-NN-ROM, and POD-DL-ROM. The baseline/enriched comparison is retained as a controlled data axis. PG/PRNN-style variants are treated as optional diagnostic tracks and are not part of the core comparative tables in this manuscript.

References

- [1] Athanasios C Antoulas. *Approximation of large-scale dynamical systems*. SIAM, 2005.
- [2] Alfio Quarteroni, Gianluigi Rozza, et al. *Reduced order methods for modeling and computational reduction*, volume 9. Springer, 2014.
- [3] Joshua Barnett, Charbel Farhat, and Yvon Maday. Neural-network-augmented projection-based model order reduction for mitigating the Kolmogorov barrier to reducibility. *Journal of Computational Physics*, 492:112420, 2023.
- [4] Albert Cohen, Charbel Farhat, Yvon Maday, and Agustin Somacal. Nonlinear compressive reduced basis approximation for PDEs. *Comptes Rendus. Mécanique*, 351(S1):357–374, 2023.
- [5] Joubine Aghili, Hassan Ballout, Yvon Maday, and Christophe Prud’Homme. Nonlinear compressive reduced basis approximation: when taylor meets kolmogorov. *arXiv preprint arXiv:2601.13712*, 2026.
- [6] Francesco Romor, Giovanni Stabile, and Gianluigi Rozza. Explicable hyper-reduced order models on nonlinearly approximated solution manifolds of compressible and incompressible navier-stokes equations. *Journal of Computational Physics*, 524:113729, 2025.
- [7] Stefania Fresca and Andrea Manzoni. Pod-dl-rom: Enhancing deep learning-based reduced order models for nonlinear parametrized pdes by proper orthogonal decomposition. *Computer Methods in Applied Mechanics and Engineering*, 388:114181, 2022.
- [8] Charbel Farhat, Todd Chapman, and Philip Avery. Structure-preserving, stability, and accuracy properties of the energy-conserving sampling and weighting method for the hyper reduction of nonlinear finite element dynamic models. *International Journal for Numerical Methods in Engineering*, 102(5):1077–1110, 2015.

- [9] S Ares De Parga, Jose Raul Bravo, JA Hernández, Rubén Zorrilla, and Riccardo Rossi. Hyper-reduction for petrov–galerkin reduced order models. *Computer Methods in Applied Mechanics and Engineering*, 416:116298, 2023.
- [10] Wenqian Chen, Qian Wang, Jan S Hesthaven, and Chuhua Zhang. Physics-informed machine learning for reduced-order modeling of nonlinear problems. *Journal of Computational Physics*, 446:110666, 2021.
- [11] Nicolas Sibuet, Sebastian Ares de Parga, Jose Raul Bravo, and Riccardo Rossi. Discrete physics-informed training for projection-based reduced-order models with neural networks. *Axioms*, 14(5):385, 2025.